

Mock Midterm, Question 1 Solutions

ECO387C Macro I, Fall 2026

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Acknowledgment. I solved every problem by hand on paper. The handwritten pages were then scanned and typeset with ChatGPT so that the LaTeX is readable, and I checked the typeset version against my notes. The numbers are checked in mock-midterm-checks.py, with output in mock-midterm-checks.txt.

$u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, $u'(c) = c^{-\gamma}$, and the transition probabilities are

$$\pi(y_h | y_h) = p_h, \quad \pi(y_l | y_h) = 1 - p_h, \quad \pi(y_l | y_l) = p_l, \quad \pi(y_h | y_l) = 1 - p_l$$

(Azzimonti et al., Section 7.1.2, p. 193).

Part (a): flow budget constraint

$$c_t + a_{t+1} = R_{t-1}a_t + y_t,$$

plus a no-Ponzi condition, which does not bind in Part (e).

Part (b): state variables

(b_t, y_t) with $b_t := R_{t-1}a_t$. y_t is income, forecasts y_{t+1} (Markov, $p_h \neq 1 - p_l$), and prices the bond, $R_t = R(y_t)$. Identical agents make the bond distribution degenerate, so no other aggregate state is needed.

Part (c): Bellman equation

With $c = b + y - a'$ and $b' = R(y)a'$,

$$V(b, y) = \max_{a'} \left\{ \frac{(b + y - a')^{1-\gamma}}{1-\gamma} + \beta \sum_{y' \in \{y_h, y_l\}} \pi(y' | y) V(R(y)a', y') \right\},$$

that is,

$$\begin{aligned} V(b, y_h) &= \max_{a'} \left\{ u(b + y_h - a') + \beta \left[p_h V(R(y_h)a', y_h) + (1 - p_h) V(R(y_h)a', y_l) \right] \right\}, \\ V(b, y_l) &= \max_{a'} \left\{ u(b + y_l - a') + \beta \left[(1 - p_l) V(R(y_l)a', y_h) + p_l V(R(y_l)a', y_l) \right] \right\}, \end{aligned}$$

as in Section 7.6, p. 219, with $q = 1/R(y)$.

Part (d): Euler equation

FOC in a' (interior), using $\partial b' / \partial a' = R(y)$,

$$-u'(b + y - a') + \beta R(y) \mathbb{E}[V_b(R(y)a', y') | y] = 0.$$

Envelope, with policy $a' = g(b, y)$,

$$\begin{aligned} V_b(b, y) &= u'(c) \left(1 - \frac{\partial g}{\partial b} \right) + \beta R(y) \mathbb{E}[V_b(b', y') | y] \frac{\partial g}{\partial b} \\ &= u'(c) + \frac{\partial g}{\partial b} \left[-u'(c) + \beta R(y) \mathbb{E}[V_b(b', y') | y] \right] = u'(c), \end{aligned}$$

the bracket being zero by the FOC. Update one period and substitute,

$$u'(c) = \beta R(y) \mathbb{E}[u'(c') | y], \quad \text{i.e.} \quad c_t^{-\gamma} = \beta R_t \mathbb{E}[c_{t+1}^{-\gamma}].$$

Part (e): market clearing and the equilibrium rate

Goods and bonds clear,

$$\int c_t = y_t, \quad \int a_{t+1} = 0.$$

Identical agents facing the same R_t choose the same a_{t+1} , so bond clearing gives $a_{t+1} = 0$, and the budget gives $c_t = y_t$ (Section 7.5, p. 217). R solves the Euler equation at $c = y$,

$$R(y) = \frac{u'(y)}{\beta \mathbb{E}[u'(y') | y]} = \frac{y^{-\gamma}}{\beta \sum_{y'} \pi(y' | y) (y')^{-\gamma}}.$$

Part (f): rates with gamma = 2

$$u'(2) = \frac{1}{4}, u'(1) = 1.$$

$$\mathbb{E}[u'(y') \mid y_h] = p_h \cdot \frac{1}{4} + (1 - p_h) \cdot 1 = \frac{p_h + 4 - 4p_h}{4} = \frac{4 - 3p_h}{4},$$

$$R_h = \frac{1/4}{\beta(4 - 3p_h)/4} = \frac{1}{\beta(4 - 3p_h)}.$$

$$\mathbb{E}[u'(y') \mid y_l] = p_l \cdot 1 + (1 - p_l) \cdot \frac{1}{4} = \frac{4p_l + 1 - p_l}{4} = \frac{1 + 3p_l}{4},$$

$$R_l = \frac{1}{\beta(1 + 3p_l)/4} = \frac{4}{\beta(1 + 3p_l)}.$$

$$R_t = \begin{cases} \frac{1}{\beta(4 - 3p_h)}, & y_t = y_h, \\ \frac{4}{\beta(1 + 3p_l)}, & y_t = y_l. \end{cases}$$

Example, $\beta = 0.95$, $p_h = 0.8$, $p_l = 0.6$,

$$R_h = \frac{1}{0.95 \cdot 1.6} = \frac{1}{1.52} = 0.658, \quad R_l = \frac{4}{0.95 \cdot 2.8} = \frac{4}{2.66} = 1.504,$$

against $1/\beta = 1.053$. For $p_h < 1$, $4 - 3p_h > 1$, and for $p_l < 1$, $1 + 3p_l < 4$, so

$$R_h < \frac{1}{\beta} < R_l.$$

Expected consumption growth is negative in a boom, so the rate that stops saving is low, and positive in a recession, so the rate is high (Figure 1).

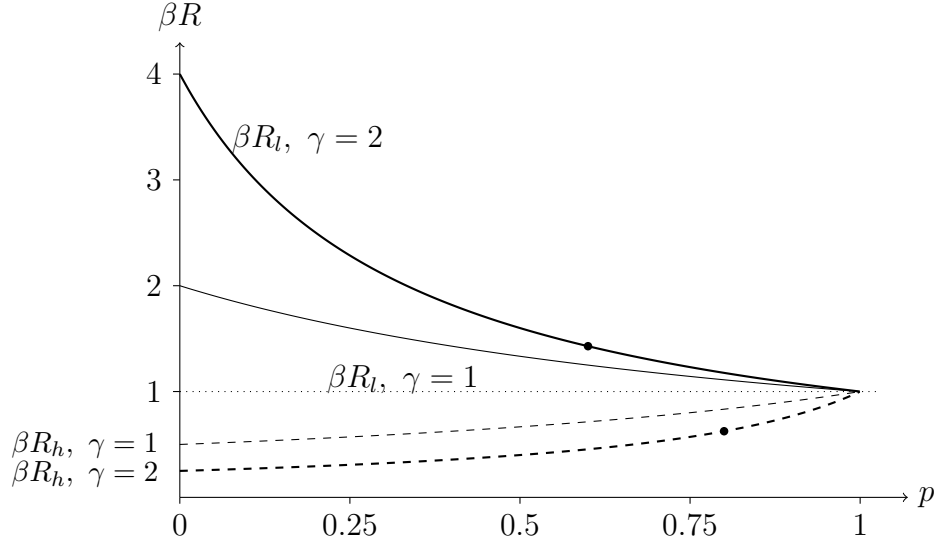


Figure 1: Equilibrium gross rate times β against the stay probability of the current state, $y_h = 2$, $y_l = 1$. Solid, recession (horizontal axis p_l). Dashed, boom (horizontal axis p_h). Thick, $\gamma = 2$. Thin, $\gamma = 1$. The dotted level is $\beta R = 1$. Dots mark the example, $p_l = 0.6$ and $p_h = 0.8$ with $\gamma = 2$.

Part (g): curvature and the iid case

The ratio of marginal utilities across states is

$$\frac{u'(1)}{u'(2)} = 2^\gamma = 2 \quad (\gamma = 1), \quad 4 \quad (\gamma = 2).$$

Higher γ (lower IES $1/\gamma$) makes the same endowment change a larger change in marginal utility, so the rate that supports $c = y$ moves further from $1/\beta$. Same numbers with log utility,

$$R_h = \frac{1}{0.95 \cdot (2 - 0.8)} = 0.877, \quad R_l = \frac{1}{0.95(0.6 + 0.4 \cdot \frac{1}{2})} = 1.316.$$

iid, $p_h = 1 - p_l$. Then $\pi(y_h | y_h) = \pi(y_h | y_l) = p_h$, so

$$\mathbb{E}[u'(y') | y_h] = \mathbb{E}[u'(y') | y_l] = p_h u'(2) + (1 - p_h) u'(1) =: m,$$

$$\frac{R_l}{R_h} = \frac{u'(1)/(\beta m)}{u'(2)/(\beta m)} = \frac{u'(1)}{u'(2)} = \frac{1}{1/4}, \quad \boxed{\frac{R_l}{R_h} = 4}$$

for any p_h, p_l, β . Example $p_h = 0.4, p_l = 0.6$: $R_h = 0.376, R_l = 1.504$.

No trade is optimal. $c_t = y_t, a_{t+1} = 0$ satisfies the budget and the Euler equation, consumption is bounded so transversality holds, and u is concave.

Sources

- Azzimonti, M., P. Krusell, A. McKay, and T. Mukoyama (2024), *Macroeconomics*, draft. Section 7.1.2 (Markov chains, p. 193), Section 7.5 (zero asset position of a representative household, p. 217), Section 7.6 (recursive consumption-savings problem, first-order and envelope conditions, Euler equation, pp. 218–219).